

Resilient Distributed Multi-Task Estimation With Network Learning Under Malicious Attacks

Yi Hua^{ID}, Fangyi Wan^{ID}, *Member, IEEE*, Youmin Zhang^{ID}, *Fellow, IEEE*, Xinlin Qing^{ID}, *Member, IEEE*,
and Hongping Gan^{ID}, *Member, IEEE*

Abstract—Network security plays a critical role in distributed estimation. When a multi-task distributed network is subjected to adversarial attacks, normal nodes are forced to both detect malicious nodes and learn cluster membership. The existing distributed algorithms, which mainly rely on some prior information and operate mechanically, remain insufficient and lack adaptability. To solve these problems, an adaptive distributed multi-task algorithm with cluster learning is proposed, which can make all normal nodes achieve resilience and secure estimation. In the proposed algorithm, an adaptive detection mechanism without any prior information is firstly proposed to identify the difference among nodes. Subsequently, a node queue strategy is designed to enable all normal nodes to achieve resilience and cluster learning. The convergence behavior of the proposed algorithm is analyzed, and then a convergence condition is derived to ensure the mean stability and mean-square convergence. Finally, based on different forms of data tampering and forgery, simulations consider three typical and distinct attacks. The results show that the proposed algorithm, without using any prior information, outperforms state-of-the-art algorithms, demonstrating the superior adaptability and effectiveness.

Index Terms—Attack intrusion, network security, adaptive detection, resilient algorithm, cluster learning, distributed estimation, secure data fusion, malicious node identification.

I. INTRODUCTION

WITH the rapid development of sensing technology, the amount of data is growing exponentially, and the demand for processing capabilities is becoming increasingly urgent [1], [2], [3]. The traditional centralized strategy falls short when dealing with large-scale data, making it difficult to meet

the real-world requirements [4]. To change this situation, the distributed strategy has become increasingly important [5]. By distributing computational tasks to networked nodes, this strategy improves data processing efficiency while enhancing system reliability and scalability, making it a key foundation for contemporary societal development [6], [7], [8].

Under the distributed strategy, nodes are interconnected to form distributed networks, which are widely applied in artificial intelligence, smart cities, and related fields [9], [10], [11], where nodes cooperate to estimate an unknown parameter of interest. Based on the least mean squares (LMS) strategy [12], [13], [14], the diffusion LMS (DLMS) algorithm has been extensively studied and widely applied in distributed networks [15], [16]. In these studies [12], [13], [14], [15], [16], the considered network is secure and normal [17]. That is, the data from all nodes are normal during communication and sensing, and can therefore be used directly. However, with the rapid development of electronic information, distributed network is not always secure [18], [19], [20], [21] and may be invaded by attacks that turn some nodes malicious [22], [23], [24], [25], [26].

When a distributed network is attacked, some nodes may sense and transmit the abnormal and malicious data [10], which can compromise the overall network performance in the absence of secure mechanisms [25]. Therefore, extensive research on network resilience against attacks has emerged, including multi-sensor fusion with Markov strategy detector [10], [25], secure distributed algorithm with Bayesian detector [26], secure distributed gradient-descent algorithm against Byzantine attacks [27], and robust DLMS algorithm with dual verification subsystems [7] against both Byzantine and manipulation attacks [28]. Differing from these works, [21] and [24] focused on false data injection (FDI) attack, which can evade certain detection mechanisms [29]. To achieve secure estimation, [21] designed a sorting-based median verification mechanism, while [24] adopted a correntropy-kernel method.

As the number of unknown interesting parameters increases, distributed network can be viewed as a multi-task network [30], [31]. In this network, those nodes that estimate the same parameter can form a cluster, which is viewed as a task. But in multi-task network, the cluster information usually is unknown in advance. Therefore, several distributed algorithms with cluster learning are developed. For example, [30] designed a partially informed agent (PIA) strategy to learn cluster, and then a multi-task DLMS with PIA (MDLMS-PIA) algorithm was proposed for multi-task estimation. Alternatively, [31] developed a preconditioned

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Yi Hua is with College Information and Artificial Intelligence, Yangzhou University, Yangzhou 225127, China, and also with the School of Aeronautics, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: yihua@mail.nwpu.edu.cn).

Fangyi Wan is with the School of Aeronautics, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: fwan@nwpu.edu.cn).

Youmin Zhang is with the Department of Mechanical, Industrial and Aerospace Engineering, Concordia University, Montreal QC H3G 1M8, Canada (e-mail: ymzhang@encs.concordia.ca).

Xinlin Qing is with the School of Aerospace Engineering, Xiamen University, Xiamen 361000, China (e-mail: xinlingqing@xmu.edu.cn).

Hongping Gan is with the School of Software, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: ganhongping@nwpu.edu.cn).

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multi-task DLMS (P-MDLMS) algorithm by approximating local information to perform clustering and estimation. However, these studies [30], [31] only focus on a single type of difference among clusters. When the network is attacked, an additional type of heterogeneity arises between normal and malicious nodes. Two types of differences can mutually influence each other, which causes a significant challenge for parameter estimation.

To the best of our knowledge, when faced with multi-task estimation under attacks, the main research can be broadly categorized into two types: one assumes that some prior information about the network is known [32]; and the other only isolates the relation between normal and malicious nodes [18], [19]. In the first category, some related multi-task information, such as task similarity or clustering structure [32], is assumed to be known a priori by the network, enabling each node to design secure tests against malicious attacks. But in practice, such prior information is often unavailable, which limits the applicability of these methods. For the second category, the network uniformly and mechanically regards other cluster nodes and the malicious nodes as different nodes according to their differences, and then classifies them into different types [18], [19]. However, the malicious nodes are only simply separated, yet still remain within the distributed network. From the perspective of the entire system, it is impossible to explicitly identify which nodes are benign and which are malicious. Moreover, to achieve resilient estimation, these algorithms [18], [19] adopt the trimmed-mean method, mechanically discarding a fixed number of nodes based on dissimilarity. Specifically, the number of nodes to be discarded needs to be predetermined, which lacks adaptivity and may also require some prior information about the attacks [28].

Therefore, from these analyses, it is concluded that the differences induced by adversarial attacks and those stemming from distinct tasks are often intertwined, severely hindering accurate detection and estimation. This entanglement poses a significant challenge for existing approaches, as most either lack practicality in real-world scenarios or fail to preserve the global structural consistency of the network. Consequently, when cluster affiliation or attack-related information is unknown, current methods for secure distributed multi-task estimation remain inadequate and insufficiently adaptive. To address these challenges, differing from them [18], [19], [32], an adaptive secure detection mechanism without any prior information is proposed to firstly identify their differences. Afterwards, a node queue strategy is designed to learn cluster information and distinguish the normal and malicious nodes. Based on the designed detection mechanism and node queue strategy, an adaptive distributed multi-task algorithm with cluster learning is proposed to enable all normal nodes to obtain network information and resilient estimation. The contributions of this paper can be included as

- 1) An adaptive secure multi-task estimation algorithm with cluster learning is proposed to learn the cluster information and identify the malicious nodes.
- 2) The designed learning mechanism and secure strategy require no prior network information, so they are suitable for the practical applications.
- 3) The convergence behaviors of the proposed algorithm are analyzed, which proves that the convergence condition can guarantee the algorithmic stability under attacks.

4) The designed strategy and algorithm will also be applied to single-task networks and various other attack scenarios.

This paper is organized as follows. In Section II, the distributed data model is formulated, and the secure multi-task estimation problem is presented. Section III proposes the designed resilient algorithm with cluster learning under attacks. Section IV analyzes the algorithm performance. Section V verifies the multi-task estimation performance under attacks by simulations, and Section VI summarizes this paper.

II. DATA MODEL AND PROBLEM FORMULATION

In this section, the distributed data model is formulated, and the secure multi-task estimation problem is subsequently derived.

A. Distributed Data Model

In this paper, a distributed network with N nodes is considered. Each node n can communicate with these nodes within its communication radius, and the set of these nodes can be denoted as $\mathcal{N}_n$ ($n \in \mathcal{N}_n$). At every time instant t , each node can sense some random process and then take a regression vector $\mathbf{u}_{n,t}$, where its dimension is $M \times 1$, and a correspondingly correlated scalar measurement $d_n(t)$. The relation of $\mathbf{u}_{n,t}$ and $d_n(t)$ [8], [12], [21], [33], [34] is expressed by the following linear formula

$$d_n(t) = \mathbf{u}_{n,t}^T \mathbf{w}_n^\circ + v_n(t), \quad (n = 1, 2, \dots, N), \quad (1)$$

where $(\cdot)^T$ denotes a transpose operation, $v_n(t)$ is an additive noise with zero mean and variance $\sigma_{v,n}^2$, and $\mathbf{w}_n^\circ$ is an unknown and ideal variable ($M \times 1$) that needs to be estimated.

For the convenience of subsequent analyses, without loss of generality, the assumption that the noise $v_n(t)$ is independently and identically distributed is introduced [12], [30]. Furthermore, any two nodes can be directly or indirectly connected through communication in the distributed network, which means that every node can diffuse its behavior and affect other nodes. Based on the relation (1) under the non-cooperation style, a private loss function $J(\mathbf{w}_{n,t})$ can be presented to estimate the unknown variable

$$J(\mathbf{w}_{n,t}) \triangleq \|d_n(t) - \mathbf{u}_{n,t}^T \mathbf{w}_{n,t}\|_2^2, \quad (2)$$

where $\mathbf{w}_{n,t}$ is the t -th estimation of $\mathbf{w}_n^\circ$. In the distributed network, each node can use the neighboring information to improve the estimation, so the local loss function $J'(\mathbf{w}_{n,t})$ with distributed cooperation can be derived as

$$J'(\mathbf{w}_{n,t}) \triangleq \sum_{j \in \mathcal{N}_n} a_{j,n} \mathbb{E} \|d_j(t) - \mathbf{u}_{j,t}^T \mathbf{w}_{n,t}\|_2^2, \quad (3)$$

where $\mathbb{E}(\cdot)$ is the expectation, and $a_{j,n}$ is the fusion weight [35]. Therefore, each node n can estimate $\mathbf{w}_n^\circ$ by the cooperative strategy.

B. Secure Multi-Task Estimation Formulation

When there are multiple different parameters that are unknown and need to be estimated, the distributed network becomes the multi-task estimation and can be divided into many clusters, which is called the clustered multi-task network [36],

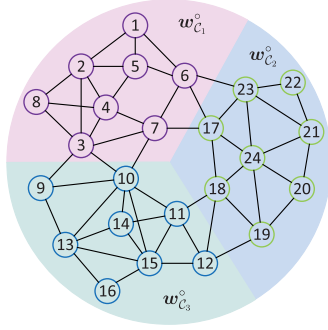


Fig. 1. The clustered multi-task network with three clusters.

according to the parameter difference. The i -th cluster is denoted as $\mathcal{C}_i$, so the network with C clusters can be viewed as the set $\{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_C\}$. As shown in Fig. 1, there are three clusters, and their estimated parameters are expressed as $w_{C_1}^o$, $w_{C_2}^o$, and $w_{C_3}^o$. Since node 1 belongs to cluster $\mathcal{C}_1$, one achieves that $w_1^o = w_{C_1}^o$.

Before the data combination, each node receives information from its neighboring nodes. However, all data that are from $\mathcal{N}_n$ may not be used directly for node n . For example, consider node 7 in Fig. 1, whose neighbor node set is $\mathcal{N}_7 = \{3, 4, 6, 7, 10, 17\}$. To obtain the accuracy estimation parameter, node 7 only fuses the information from $\mathcal{C}_1$, and thus does not aggregate information from nodes 10 and 17 that belong to $\mathcal{C}_2$ and $\mathcal{C}_3$, respectively. In general, the prior cluster information is unknown to each node. Therefore, each node n should identify the cluster membership of its neighbors before fusing their information. That is, if $n \in \mathcal{C}_i$, node n should learn the cluster information $\mathcal{C}_i$.

When the distributed network is invaded by attackers, some nodes become malicious. In the secure distributed estimation, to guarantee the resilience of node, the network should satisfy an f -model [18], [19], [21], [32]:

Definition 1: (f -model) A network is f -model, if all nodes satisfy: Up to f neighbors are attacked among $|\mathcal{N}_n|$ neighbors.

For this model, in the event of a significant increase in the magnitude of f , the node encounters challenges in accessing reliable information essential for establishing resilience in the absence of any a priori knowledge. Therefore, in promising that the distributed network can be meaningfully recovered and resilient under attacks, the worst situation considered is that f does not exceed $|\mathcal{N}_n|/2$ [8], [21], [27], [32]. This means that if all normal nodes can obtain the resilient estimation, every normal node owns more than half of honest and same-cluster neighbors so that a reliable median is available to provide detection, which can be called mid-value method [21], [27], [37], [38]. However, in a multi-task network, some nodes will communicate with other clustered nodes, so there is a universal situation that some honest neighbors belong to the different cluster, which makes the median unreliable and causes their secure algorithms with mid-value method malfunction.

Similarly [18], [19], [21] and [32], in this paper considering the multi-task network, the f -model is first satisfied, and the f attacked neighbors of node n are assumed to satisfy $2f + 1 \leq |\mathcal{N}_n|$. However, in multi-task settings, satisfying the f -model alone is not sufficient to guarantee secure estimation,

since a normal node may have fewer same-cluster neighbors than the required majority, even when the global percentage of compromised nodes is small. The differences caused by adversarial attacks and those resulting from distinct tasks are often intertwined, severely hindering accurate detection and identification. Yet, to the best of our knowledge, this critical issue has received little attention and remains underexplored in existing studies. Considering all these problems, a secure detection mechanism operates without any prior information to identify the differences among nodes. Based on this detection, a node queue strategy is designed to learn cluster information and enhance the adaptability and security of the multi-task network. Through the designed methods, the proposed algorithm can get a resilient estimation, which will be more compatible with multi-task network and can protect the entire network from the impact of malicious nodes.

III. RESILIENT DISTRIBUTION MULTI-TASK ALGORITHM WITH CLUSTER LEARNING UNDER ATTACKS

To achieve resilient distributed multi-task estimation with cluster learning, the proposed secure algorithm consists of four components: *adaptation*, *communication*, *node queue*, and *resilient combination*. These components are executed once per sample [39]. In this section, the overall procedures of these components are introduced, and then the queue method will be explained in detail for data detection and resilience.

In step of *adaptation*, combined the local loss function (3) with the gradient descent method, a negative gradient update equation can be got for each node

$$\psi_{n,t} = w_{n,t} + \mu_n \mathbf{u}_{n,t} (d_n(t) - \mathbf{u}_{n,t}^T w_{n,t}), \quad (4)$$

where $\psi_{n,t}$ is the intermediate estimation, and μ_n is step-size.

In step of *communication*, each node n should communicate with its neighboring nodes, and exchange the intermediate estimation $\psi_{n,t}$.

In step of *node queue*, each node uses the detection mechanism to test the node state. Afterwards, based on this state, node n chooses to start queuing operation under the currently available information, which will be clarified in Section III-B. Therefore, each node can obtain two sets: the secure and same-cluster neighbor set $\mathcal{N}_n^s \cap \mathcal{C}_i$ (assuming $n \in \mathcal{C}_i$) and other neighbor set $\mathcal{N}_n^o$.

Remark 1: Noting the set $\mathcal{N}_n^o$, there may be three types of nodes. The first type is obviously other clusters that may include the both malicious and normal nodes and must be isolated by node n . The second type is the same cluster but malicious. The last type is the same cluster and normal. At the current situation, the last type of node may not start queuing operation to obtain their state information, so their information cannot be fused at present. But, as the node queue strategy under the gradual iteration, the last type of node will be absorbed gradually in set $\mathcal{N}_n^s \cap \mathcal{C}_i$.

In step of *resilient combination*, each node n calculates the adaptive fusion weight $a_{j,n}(t)$ according to set $\mathcal{N}_n^s \cap \mathcal{C}_i$, and then updates the estimation $w_{n,t}$. When node n is secure and

normal, the resilient combination will be

$$\mathbf{w}_{n,t+1} = \sum_{j \in \mathcal{N}_n^s \cap \mathcal{C}_i} a_{j,n}(t) \psi_{j,t}. \quad (5)$$

According to Remark 1, it is obvious that the first type must be isolated by the following detection mechanism, so the latter two situations should be mainly considered. The third type is normal but cannot start queuing operation, so it is better not to combine other variables in order to maintain its stable performance. Since the second type is malicious, it also does not combine other variables, which can provide the detection strategy of Section III-A. Hence, when node n belongs to the latter two types, it updates itself by non-cooperation manner

$$\mathbf{w}_{n,t+1} = \psi_{n,t}. \quad (6)$$

Therefore, through the iterations of four steps, the cluster information can be learned, and the secure distributed multi-task estimation will be also achieved.

A. The Detection Mechanism

When each node n has sent its $\psi_{n,t}$ and received the $\psi_{j,t}$ from $\mathcal{N}_n$, it should check the state of neighboring information. To distinguish the difference among nodes, the distance $e_{n,j}(t)$ between node n with its neighbors can be tested as

$$e_{n,j}(t) = \|\psi_{n,t} - \psi_{j,t}\|_2 \leq \alpha, \quad (j \in \mathcal{N}_n), \quad (7)$$

where α is a threshold. If formula (7) holds, the data $\psi_{j,t}$ can be viewed as the same cluster with node n ; otherwise, they are different. Noting formula (7), the threshold α needs to be predefined properly according to $e_{n,j}(t)$, and is also highly sensitive to the noise statistics [40]. If α is set randomly, the detection will reduce the estimation performance. Furthermore, it is clear that α belongs to $(0, +\infty)$. Therefore, α should be set empirically and cannot be searched directly.

To improve the above situation, combining with the basic Geman-McClure function [41], an adaptive detection form with the adjusting factor z is designed as following equation

$$g_{n,j}(t) \triangleq \frac{1}{\left(1 + \left(\frac{e_{n,j}(t)}{z}\right)^2\right)^2} \geq \alpha', \quad (j \in \mathcal{N}_n), \quad (8)$$

where $\alpha' \in (0, 1)$ is a threshold that is different from α , and $z \in (0, +\infty)$ denotes a differential adjustment factor. It is obvious that α' reflects this distance $e_{n,j}(t)$ and quantifies their similarity, and z can adjust the degree of similarity. When the distance $e_{n,j}(t)$ is relatively small, the value of z is typically set within the range $(0, 1)$ to enhance similarity and improve detection performance. However, when the distance $e_{n,j}(t)$ is relatively large, z should be set within the range $(1, +\infty)$ to adjust the difference in distances, which facilitates the selection of an appropriate threshold α' and thereby leads to better detection performance. Noting (8), when node j becomes more similar to node n , $g_{n,j}(t)$ will be closer to 1; otherwise, $g_{n,j}(t)$ is closer to 0.

For the sake of expressions, some notions based on multi-task network can be introduced as follows

Definition 2: (Border node) As long as there is a neighboring node j that makes $g_{n,j}(t) < \alpha'$, node j should be regarded as the different state from node n , so node n is defined as the border node.

Definition 3: (Inner node) When all neighboring nodes satisfy detection (8), node n is considered as the inner node.

Definition 4: (f -local node) In multi-task network, if node n is f -local node, at least half or more of its neighbors share the same state as node n itself.

Remark 2: In distributed multi-task network with f -model, all distributed algorithms related with the mid-value method require that all nodes must be f -local node, which may be impossible. In multi-task scenarios, a normal node often has a limited number of same-cluster neighbors, making it difficult or even impossible to satisfy the f -local node. Differing from mid-value method, in this paper, it is not required that all nodes satisfy this Definition 4, which can better reflect the realities of the world. Because node n communicates with other clusters, it will cause node n to not fit Definition 4 even if only one of its neighbor is attacked [18], [19], [21], [32]. Therefore, a node may still satisfy the f -model in a global sense while failing to obtain any reliable majority information locally, breaking the fundamental assumption of mid-value-based estimators. For example, in Fig. 1, when node 5 is attacked, node 6 is not an f -local node but satisfies f -model. However, node 6 may not obtain the reliable median to provide the resilience.

According to Definition 1, it is obvious that if a node is f -local node, this node is resilient, which means that it can obtain the normal and honest estimation. Combining with (8), a set $\mathcal{N}_n^a$ that agrees with node n can be got as

$$\mathcal{N}_n^a \triangleq \{j \in \mathcal{N}_n | g_{n,j}(t) \geq \alpha'\}. \quad (9)$$

Observing these notations, it is concluded that the inner node is $|\mathcal{N}_n^a| = |\mathcal{N}_n|$, while the border node brings $|\mathcal{N}_n^a| \neq |\mathcal{N}_n|$. In multi-task network with attacks, we can find that the other cluster node and the malicious node can cause different states from node n , which makes node n become the border node. Therefore, it is easy to focus on the border node in the following discussion. For border node, when $|\mathcal{N}_n^a| \geq |\mathcal{N}_n|/2$, node n is f -local node. But if $|\mathcal{N}_n^a| < |\mathcal{N}_n|/2$, it is not easily identified. To distinguish this situation, the following node queue strategy can be designed based on (9).

B. Node Queue Strategy

By aforementioned analyses, it is known that a distributed network ensures that all normal nodes should be the f -local node, but it is impossible to every node in multi-task network. Combining with Definition 4, another notion, which is more suitable to the realistic applications, is introduced

Definition 5: (Reachable node) If node n is the reachable node, it should satisfy:

- There is a communication path connected with node n , and all nodes of this path have the same estimation state;
- At least one of these nodes is the f -local node.

Remark 3: In this paper, the considered network is a clustered multi-task network (as shown in Fig. 1) [36]. Therefore, the

situation of Definition 5 is frequently encountered. Moreover, the condition for reachable nodes is naturally fulfilled in the clustered multi-task networks.

To make all normal nodes obtain resilient estimation in a multi-task network, it requires that each normal node should be the reachable node. Compared with the f -local node, the reachable node would be relaxed and common in multi-task situation. Meanwhile, it is evident that an f -local node must be a reachable node, but the reverse is not always true. In the multi-task network, to identify the border node, the following variable can be introduced

$$s_n(t) = \begin{cases} 1, & \text{if } |\mathcal{N}_n^a| \geq |\mathcal{N}_n|/2; \\ 0, & \text{otherwise,} \end{cases} \quad (10)$$

where the variable $s_n(t)$ denotes the state of f -local node.

When $s_n(t) = 1$, the node n has determined its state. Therefore, it can be considered that node n can make a queue stand to itself, and the queue variable $q_n(t)$ can be set as 1. When $s_n(t) = 0$, there exists a node j in order that $s_j(t) = 1$ ($j \in \mathcal{N}_n^a$), and then $q_n(t) = 1$; otherwise, $q_n(t) = 0$. Based on (9), it is clear that $n \in \mathcal{N}_n^a$ holds, so the above situations can be uniformly concluded as

$$q_n(t) = \begin{cases} 1, & \text{if } \exists s_j(t) = 1; \\ 0, & \text{otherwise,} \end{cases} \quad (11)$$

where $j \in \mathcal{N}_n^a$. According to (11), $q_n(t) = 1$ denotes that there is at least one node j whose state information is definitely determined, and which is in the same state as node n , so that node n is queued up to node j .

To improve the performance of this queue (11), a smooth process can be proposed as follows

$$Q_n(t) = \nu Q_n(t-1) + (1-\nu)q_n(t), \quad (12)$$

where $\nu \in (0, 1)$ is the strictly convex combination coefficient, and $Q_n(t)$ denotes a trust level of queue variable. By the trust level $Q_n(t)$, the following binary hypothesis test is obtained

$$Q_n(t) \underset{H_1}{\overset{H_0}{\geq}} \gamma, \quad (13)$$

where $\gamma \in (0, 1)$ is a test threshold that can be configured by a traversal search, and H_0 and H_1 represent the hypotheses about whether node n can queue with a node from $\mathcal{N}_n^a$.

Noting (11), it is found that the queue information is merely transmitted between the direct neighbors, which means that the reachable node can not completely obtain the resilience. Besides, when H_0 holds in (13), it indicates that this node already has a queue. That is, node that has already been enqueued is restricted from disseminating their queue-related information to other reachable nodes. Therefore, in return, combining with the test (13), the queue (11) can be further adjusted as the following node queue strategy

$$q_n(t) = \begin{cases} 1, & \text{if } \exists s_j(t) = 1 \text{ or } \exists Q_j(t-1) \geq \gamma; \\ 0, & \text{otherwise,} \end{cases} \quad (14)$$

where $j \in \mathcal{N}_n^a$. By the node queue strategy (14), the reachable node would be absorbed and then queued by f -local node in its

Algorithm 1: Resilient Distributed MDLMS-NQ Algorithm.

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Initialize  $\psi_{n,0} = \mathbf{w}_{n,0} =$  zero vector, and configure  $\mu_n$ ,
 $\alpha'$ ,  $z$ ,  $\nu$  and  $\gamma$ 
for  $t \geq 0$  do
  for  $n = 1, 2, \dots, N$  do
     $\psi_{n,t} = \mathbf{w}_{n,t} + \mu_n \mathbf{u}_{n,t}(d_n(t) - \mathbf{u}_{n,t}^T \mathbf{w}_{n,t})$ ;
    for  $j \in \mathcal{N}_n$  do
      compute the  $g_{n,j}(t)$  by (8);
      obtain the set  $\mathcal{N}_n^a$  by (9);
      identify the state variable  $s_n(t)$  by (10);
      update the queue variable  $q_n(t)$  by (14);
    end for
    get the trust level  $Q_n(t)$  by (12);
    if  $H_0$  of (13) holds
      clarify the set  $\mathcal{N}_n^a$  by (15);
      alter weight  $a_{j,n}(t)$  by (17);
       $\mathbf{w}_{n,t+1} = \sum_{j \in \mathcal{N}_n^s \cap \mathcal{C}_i} a_{j,n}(t) \psi_{j,t}$ ;
    else  $H_1$  of (13) holds
      clarify the set  $\mathcal{N}_n^a$  by (16);
       $\mathbf{w}_{n,t+1} = \psi_{n,t}$ ;
    end if
  end for
end for

```

communication path. Therefore, the last type of Remark 1 can obtain the resilient estimation.

Based on (13), the set $\mathcal{N}_n^a$ can be clarified as: if H_0 holds, this node obtains the resilient combination (5) by

$$\mathcal{N}_n^s \cap \mathcal{C}_i = \mathcal{N}_n^a, \text{ and } \mathcal{N}_n^o = \mathcal{N}_n \setminus \mathcal{N}_n^a; \quad (15)$$

otherwise H_1 holds, this node adopts non-cooperation update (6) by

$$\mathcal{N}_n^s \cap \mathcal{C}_i = \emptyset, \text{ and } \mathcal{N}_n^o = \mathcal{N}_n. \quad (16)$$

By the above detection mechanism and node queue strategy, all normal nodes will learn the cluster information and identify the honest neighbors. Noting (5), when two sets $\mathcal{N}_n^s \cap \mathcal{C}_i$ and $\mathcal{N}_n^o$ have been achieved by (15), the weight $a_{j,n}(t)$ can be obtained by the following adaptive Metropolis rule

$$a_{j,n}(t) = \begin{cases} 1/\max(m_{n,t}, m_{j,t}), & \text{if } j \in \mathcal{N}_n^s \cap \mathcal{C}_i; \\ 1 - \sum_{k \in \mathcal{N}_n^s \cap \mathcal{C}_i} a_{k,n}(t), & \text{if } j = n; \\ 0, & \text{otherwise,} \end{cases} \quad (17)$$

where $\mathcal{N}_n^s \triangleq \mathcal{N}_n^s \setminus \{n\}$, and $m_{n,t} = |\mathcal{N}_n^s \cap \mathcal{C}_i|$. Therefore, the proposed resilient distributed multi-task DLMS algorithm with node queue strategy (MDLMS-NQ) can be got, and all procedures can be concluded in Algorithm 1, where parameters μ_n , α' , z , ν and γ are defined and obtained easily that will be verified by simulation of Section V.

IV. PERFORMANCE ANALYSIS

According to the proposed MDLMS-NQ algorithm, normal and malicious nodes exhibit distinct data update behaviors. Therefore, this section first analyzes the mathematical behavior

of malicious nodes, followed by the convergence performance of normal nodes. Finally, the detection performance and computational complexity are evaluated.

Remark 4: The proposed algorithm considers multiple attacks (detailed in Section V), which cause different deviations. However, from the perspective of attackers, their goal is always to disrupt the estimation $\mathbf{w}_{n,t}$ in the distributed network. Therefore, despite the variety of attacks, the performance analysis can focus on the behavior of $\mathbf{w}_{n,t}$ under these adversarial conditions.

To make the analysis clear, the different error forms are introduced for malicious and normal nodes. For malicious node, the estimations $\psi_{n,t}$ and $\mathbf{w}_{n,t}$ will converge to a bad value that is different from $\mathbf{w}_n^\circ$ and could be defined as $\mathbf{w}_n^{\circ'}$. Therefore, for malicious node, the following error variables can be introduced

$$\hat{\psi}_{n,t}^a = \mathbf{w}_n^{\circ'} - \psi_{n,t}, \text{ and } \hat{\mathbf{w}}_{n,t}^a = \mathbf{w}_n^{\circ'} - \mathbf{w}_{n,t}. \quad (18)$$

Differing from (18), for normal node, two other error variables are introduced

$$\hat{\psi}_{n,t} = \mathbf{w}_n^\circ - \psi_{n,t}, \text{ and } \hat{\mathbf{w}}_{n,t} = \mathbf{w}_n^\circ - \mathbf{w}_{n,t}. \quad (19)$$

A. Mathematical Behavior of Malicious Node

Combining two errors (18) with data relation (1), we have

$$\hat{\mathbf{w}}_{n,t+1}^a = (I_M - \mu_n \mathbf{U}_{n,t}) \hat{\mathbf{w}}_{n,t}^a + \mu_n \mathbf{u}_{n,t} v_n(t), \quad (20)$$

where I_M denotes the unit matrix with dimension $M \times M$, and $\mathbf{U}_{n,t} \triangleq \mathbf{u}_{n,t} \mathbf{u}_{n,t}^\top$. By the expectation operator $E(\cdot)$ and $\mathbf{U}_n = E(\mathbf{U}_{n,t})$, (20) can be written as

$$E(\hat{\mathbf{w}}_{n,t+1}^a) = (I_M - \mu_n \mathbf{U}_n) E(\hat{\mathbf{w}}_{n,t}^a). \quad (21)$$

Squaring both sides of (20), one yields

$$\begin{aligned} \|\hat{\mathbf{w}}_{n,t+1}^a\|^2 &= ((I_M - \mu_n \mathbf{U}_{n,t}) \hat{\mathbf{w}}_{n,t}^a)^\top (\mu_n \mathbf{u}_{n,t} v_n(t)) \\ &\quad + (\mu_n \mathbf{u}_{n,t} v_n(t))^\top ((I_M - \mu_n \mathbf{U}_{n,t}) \hat{\mathbf{w}}_{n,t}^a) \\ &\quad + \|(I_M - \mu_n \mathbf{U}_{n,t}) \hat{\mathbf{w}}_{n,t}^a\|^2 + \|\mu_n \mathbf{u}_{n,t} v_n(t)\|^2. \end{aligned} \quad (22)$$

Taking expectation for (22), one achieves

$$\begin{aligned} E\|\hat{\mathbf{w}}_{n,t+1}^a\|^2 &= E\|(I_M - \mu_n \mathbf{U}_{n,t}) \hat{\mathbf{w}}_{n,t}^a\|^2 \\ &\quad + E\|\mu_n \mathbf{u}_{n,t} v_n(t)\|^2. \end{aligned} \quad (23)$$

An operation notation is introduced that $\|X\|_Y^2 \triangleq X^\top Y X$ for X and Y with suitable dimensions. Therefore, (23) is equivalent to

$$E\|\hat{\mathbf{w}}_{n,t+1}^a\|^2 = E\|\hat{\mathbf{w}}_{n,t}^a\|_\Theta^2 + \mu_n^2 \sigma_{v,n}^2 E(\mathbf{u}_{n,t}^\top \mathbf{u}_{n,t}), \quad (24)$$

where $\Theta \triangleq (I_M - \mu_n \mathbf{U}_n)^\top (I_M - \mu_n \mathbf{U}_n)$. Combining with the trace operation $\text{TR}(\cdot)$ of matrix, (24) is to become

$$E\|\hat{\mathbf{w}}_{n,t+1}^a\|^2 = E\|\hat{\mathbf{w}}_{n,t}^a\|_\Theta^2 + \mu_n^2 \sigma_{v,n}^2 \text{TR}(\mathbf{U}_n). \quad (25)$$

By operator $\text{Vec}(\cdot)$ [12], which stacks the columns of its matrix argument on top of each other, when $t \rightarrow \infty$, the mean square deviation (MSD) can obtain from (25) as follows

$$E\|\hat{\mathbf{w}}_{n,\infty}^a\|_\tau^2 = \mu_n^2 \sigma_{v,n}^2 \text{Vec}(\mathbf{U}_n)^\top (I_{M^2} - \Theta')^{-1} \tau, \quad (26)$$

where $\Theta' = (I_M - \mu_n \mathbf{U}_n)^\top \otimes (I_M - \mu_n \mathbf{U}_n)^\top$, $\tau = \text{Vec}(I_M)$, and $\otimes$ denotes Kronecker operation [21], [42], [43].

Therefore, the mean relation and mean square relation of malicious node can be derived as (21) and (26). To make analysis easy, a following lemma [12], [18], [19] can be got

Lemma 1: For each node n , when the step-size μ_n satisfies the condition (27), where $\lambda_{\max}(\cdot)$ is the maximum eigenvalue, the matrix $I_M - \mu_n \mathbf{U}_n$ will be stable.

$$0 < \mu_n < \frac{2}{\max \lambda_{\max}(\mathbf{U}_j)}, \quad (j = 1, 2, \dots, N). \quad (27)$$

Proof: If a matrix needs to be stable, its spectral radius $\rho(\cdot)$ is smaller than 1. Therefore, if condition $\rho(I_M - \mu_n \mathbf{U}_n) < 1$ is satisfied, $I_M - \mu_n \mathbf{U}_n$ will be stable. When all eigenvalues $\lambda(I_M - \mu_n \mathbf{U}_n)$ lie within $(-1, 1)$, the spectral radius condition can be satisfied. Based on this, an eigenvalue condition $-1 < 1 - \mu_n \lambda(\mathbf{U}_n) < 1$ can be achieved. It is clear that as if the sufficient condition (27) holds, the above condition must hold. The proof is completed. ■

When the matrix $I_M - \mu_n \mathbf{U}_n$ is stable, the estimation error $\hat{\mathbf{w}}_{n,t}^a$ in formula (21) will become zero as $t \rightarrow \infty$. Besides, if Θ' is stable, the MSD (26) can converge. It is found that the stability of Θ' is up to $I_M - \mu_n \mathbf{U}_n$. Based on Lemma 1, it is concluded that when step-size μ_n is sufficiently small to satisfy the condition (27), the stability of $I_M - \mu_n \mathbf{U}_n$ can be guaranteed. That is, the mean stability and mean square convergence can be got to malicious node. Therefore, when step-size μ_n is small enough, the malicious node invaded by attackers will make its estimation $\mathbf{w}_{n,t}$ converge to $\mathbf{w}_n^{\circ'}$.

B. Convergence Behavior of Normal Node

The following analysis is conducted under the condition that the node set has been correctly identified as $\mathcal{N}_n^s \cap \mathcal{C}_i$. From (19) and (1), the estimation error of normal node can be derived as

$$\begin{aligned} \hat{\psi}_{n,t} &= \hat{\mathbf{w}}_{n,t} - \mu_n \mathbf{u}_{n,t} (\mathbf{u}_{n,t}^\top \hat{\mathbf{w}}_{n,t} - v_n(t)); \\ \hat{\mathbf{w}}_{n,t+1} &= \sum_{j \in \mathcal{N}_n^s \cap \mathcal{C}_i} a_{j,n}(t) \hat{\psi}_{j,t}. \end{aligned} \quad (28)$$

The (28) can be rewritten as

$$\begin{aligned} \hat{\mathbf{w}}_{n,t+1} &= \sum_j a_{j,n}(t) (I_M - \mu_j \mathbf{U}_{j,t}) \hat{\mathbf{w}}_{j,t} \\ &\quad + \sum_j a_{j,n}(t) \mu_j \mathbf{u}_{j,t} v_j(t). \end{aligned} \quad (29)$$

where $j \in \mathcal{N}_n^s \cap \mathcal{C}_i$. Taking expectation for (29), one yields

$$E(\hat{\mathbf{w}}_{n,t+1}) = \sum_j a_{j,n}(t) (I_M - \mu_j \mathbf{U}_j) E(\hat{\mathbf{w}}_{j,t}). \quad (30)$$

Similarly, from (29), the mean square relation of normal node can be denoted as

$$E\|\hat{\mathbf{w}}_{n,t+1}\|^2 = E\|\hat{\mathbf{w}}_{j,t}\|_\Phi^2 + E\left\| \sum_j a_{j,n}(t) \mu_j \mathbf{u}_{j,t} v_j(t) \right\|^2, \quad (31)$$

where $\Phi \triangleq \Psi^T \Psi$, $j \in \mathcal{N}_n^s \cap \mathcal{C}_i$ and $\Psi = \sum_j a_{j,n}(t)(I_M - \mu_j \mathbf{U}_j)$. Combining the trace operation with the property of noise $v_n(t)$, (31) is rewritten as

$$\mathbb{E} \|\hat{\mathbf{w}}_{n,t+1}\|^2 = \mathbb{E} \|\hat{\mathbf{w}}_{j,t}\|_{\Phi}^2 + \sum_j a_{j,n}^2(t) \mu_j^2 \sigma_{v,j}^2 \text{TR}(\mathbf{U}_j). \quad (32)$$

When $t \rightarrow \infty$ under vectorization and $\Phi' = \Psi^T \otimes \Psi^T$, MSD of normal node can be obtained [12], [21], [44] as

$$\mathbb{E} \|\hat{\mathbf{w}}_{n,\infty}\|_{\tau}^2 = \sum_j a_{j,n}^2(t) \mu_j^2 \sigma_{v,j}^2 \text{Vec}(\mathbf{U}_j)^T (I_{M^2} - \Phi')^{-1} \tau. \quad (33)$$

Observing (21) and (26) of malicious node, the formulas (30) and (33) of normal node are in the form of weighted sums. Noting the weight $a_{j,n}(t)$ according to (17), it is easily found that, for node n at any time t , all its corresponding weights must be non-negative and satisfy

$$a_{1,n}(t) + a_{2,n}(t) + \dots + a_{N,n}(t) = 1. \quad (34)$$

According to the convex combination (34), the stability of Ψ is up to $I_M - \mu_j \mathbf{U}_j$ ($j \in \mathcal{N}_n^s \cap \mathcal{C}_i$). Therefore, when all nodes make their step satisfy (27), the matrix Ψ will be stable.

When Ψ is stable, (30) will be also stable, which means that the normal node can obtain the mean stability. Similarly, the formula $\sum_j a_{j,n}^2(t) \mu_j^2 \sigma_{v,j}^2 \text{Vec}(\mathbf{U}_j)^T (I_{M^2} - \Phi')^{-1} \tau$ is decided by $I_{M^2} - \Phi'$, and then $I_{M^2} - \Phi'$ is up to Φ' . When the stability of Ψ can be got, Φ' is also stable. That is, the step (5) can make normal node converge. Therefore, the mean stability and mean square convergence of normal node are also obtained.

C. Analysis of the Detection Performance

In this part, two types of detection scenarios are analyzed: distinguishing normal nodes from malicious nodes, and differentiating normal nodes belonging to different clusters. If they are identified perfectly, then each normal node can learn the cluster information and distinguish all bad nodes.

According to the aforementioned analyses, it is clear that for each node n , when step-size μ_n is small enough and satisfies the condition (27), all normal and malicious nodes can obtain the mean stability and mean-square convergence. Although they have the same condition (27) to obtain performance of stability and convergence, the different convergence situations can be obtained. The normal node will be $\mathbf{w}_n^o$, while malicious node will converge to a different value, denoted by $\mathbf{w}_n^{o'}$. Besides, from (26), the mean-square error of malicious node would be $\mu_n^2 \sigma_{v,n}^2 \text{Vec}(\mathbf{U}_n)^T (I_{M^2} - \Theta')^{-1} \tau$. But for normal node based on (33), its mean-square error is approximately $\sum_{j \in \mathcal{N}_n^s \cap \mathcal{C}_i} a_{j,n}^2(t) \mu_j^2 \sigma_{v,j}^2 \text{Vec}(\mathbf{U}_j)^T (I_{M^2} - \Phi')^{-1} \tau$. Observing the two mean-square errors, it is easy that they are the higher order infinitesimal about μ_n , which means that when μ_n is very small, they will approximate $\mathbf{w}_n^{o'}$ or $\mathbf{w}_n^o$ with small fluctuations, respectively. For attackers, it must make $\mathbf{w}_n^{o'}$ differ from $\mathbf{w}_n^o$, because their deliberate aim is to compromise the accuracy of estimation. Note that $g_{n,j}(t)$ of (8) is based on $e_{n,j}(t)$, which means $g_{n,j}(t)$ can detect the difference between $\mathbf{w}_n^{o'}$ and $\mathbf{w}_n^o$ under the suitable threshold α' and adjusting

factor z . Therefore, the proposed method and algorithm can detect the malicious nodes.

For two normal nodes n and k that are assumed to belong to different clusters, as $t \rightarrow \infty$, it has $\mathbf{w}_{n,t} \rightarrow \mathbf{w}_n^o$ and $\mathbf{w}_{k,t} \rightarrow \mathbf{w}_k^o$. Meanwhile, when condition (27) is satisfied, their mean-square errors are $\sum_j a_{j,n}^2(t) \mu_j^2 \sigma_{v,j}^2 \text{Vec}(\mathbf{U}_j)^T (I_{M^2} - \Phi')^{-1} \tau$, where j belongs to the corresponding clusters, respectively. Similarly, their mean-square errors are also the higher order infinitesimal about step-size. That is, there must always be a difference between $\mathbf{w}_{n,t}$ and $\mathbf{w}_{k,t}$, which means that $\psi_{n,t}$ and $\psi_{k,t}$ are also different when they are normal. Hence, $g_{n,k}(t)$ of (8) can identify the difference and distinguish the estimations from the different clusters, and the cluster information will be learned. When the false alarm occurs, according to Remark 1, normal node n may be mistakenly considered as a bad node, so node n updates itself following (6). Combined with the analyses in Section IV-A and Section IV-B, although node n has the same mean-square error as the attacked nodes, they converge to $\mathbf{w}_n^o$ and $\mathbf{w}_n^{o'}$, respectively. Based on Remark 4, two convergence values will inevitably differ. Therefore, after several iterations, the queue strategy can gradually correct it back to the good node. When the missed detection occurs, the information of node n is continuously corrupted and treated as a false bad node. According to the proposed algorithm, node n updates itself following (6), which implies that the erroneous information introduced by the missed detection is isolated by node n . Subsequently, node n gradually recovers to a normal estimate and then re-enters the queue strategy, being correctly absorbed by the normal nodes. Hence, the condition assumption of the node set $\mathcal{N}_n^s \cap \mathcal{C}_i$ in Section IV-B is reasonable. As such, the analysis mainly focuses on detection performance of the proposed node queue strategy and secure estimation.

Based on the above analyses, it is concluded that the proposed algorithm can isolate the malicious nodes and learn the cluster information without any prior network information to estimate the unknown parameter.

D. Complexity Analysis

To thoroughly verify estimation performance under attacks without cluster information, some related algorithms will be compared in simulations: MDLMS-PIA algorithm [30], P-MDLMS algorithm [31], reputation DLMS (RDLMS) algorithm [45], secure DLMS (SDLMS) algorithm [21], and resilient multi-task diffusion least mean Geman-McClure estimator (RMDLMG) algorithm [19] (its corresponding trimmed-mean parameter F is set as 1 or 2, respectively). Because those related algorithms only consider a single type of difference, some malicious nodes may not be effectively identified. To facilitate analyses, let K denote the number of averaged neighbors for each node, ignoring network attacks here. The impact of the attacks will be verified through the following experiments. Time and space complexities are analyzed in each iteration, shown in Table I. The proposed MDLMS-NQ algorithm has the competitive time complexity and a slightly higher space complexity. Because MDLMS-NQ algorithm simultaneously identifies two types of differences, it achieves more robust and accurate parameter estimation than others.

TABLE I
TIME COMPLEXITY AND SPACE COMPLEXITY

Algorithms	Time complexity	Space complexity
MDLMS-PIA [30]	$\mathcal{O}(NK)$	$\mathcal{O}(NM + N^2)$
P-MDLMS [31]	$\mathcal{O}(NK + NM^2)$	$\mathcal{O}(NM^2)$
RDLMS [45]	$\mathcal{O}(NK \log_2 K)$	$\mathcal{O}(NM)$
SDLMS [21]	$\mathcal{O}(NK \log_2 K)$	$\mathcal{O}(NM)$
RMDLMG [19]	$\mathcal{O}(NK)$	$\mathcal{O}(NM + N^2)$
MDLMS-NQ	$\mathcal{O}(NK)$	$\mathcal{O}(NM + N^2)$

V. SIMULATION PERFORMANCE VERIFICATION

In this section, three representative attacks of different types are considered, each involving distinct forms of data tampering and forgery [46], [47]. The estimation performance of the proposed algorithm, along with that of related algorithms, is validated through different experiments.

A. Malicious Attacks

Combining the data relation (1) with the exchanged data format, in simulations, three different types of attacks are tested, namely, Byzantine attack, compromised communication attack, and FDI attack. These attacks can tamper with the different data to disrupt the estimation. Their specific forms are described as follows.

For Byzantine attack [45], it can invade node n and then sabotage its measurement data $d_n(t)$ by

$$d_n(t) \rightarrow d_n^m(t), \quad (35)$$

where $d_n^m(t)$ is an arbitrary value [38] and can arbitrarily deviate from the rule of data relation (1).

For the compromised communication attack [21], [48], it can destroy the communication path between two nodes. In this paper, if the compromised communication attack occurs in node n , it is assumed that all its communicated data will be tampered with, which differs from [48]. That is, node n will send the malicious data to all neighboring nodes by the following mode

$$\psi_{n,t} \rightarrow \psi_{n,t}^m, \quad (36)$$

where $\psi_{n,t}^m$ is a malicious variable. In the situation, all communication paths of node n are considered to be compromised. This is the worst situation that we consider. Besides, if a secure algorithm can detect this worst situation, it must work in the situation [48] that some of the communication paths are compromised.

For FDI attack, it can tamper with the data and evade some detection methods [8], [21] by the following elaborate model

$$\begin{aligned} d_n^m(t) &= d_n(t) + d_n^{\text{FDI}}(t); \\ \mathbf{w}_{n,t}^m &= \mathbf{w}_{n,t} + \mathbf{w}_{n,t}^{\text{FDI}}, \end{aligned} \quad (37)$$

where $d_n^m(t)$ and $\mathbf{w}_{n,t}^m$ are the malicious data, and $d_n^{\text{FDI}}(t)$ and $\mathbf{w}_{n,t}^{\text{FDI}}$ are the injected bad data, which are manipulated by FDI attack and satisfy $d_n^{\text{FDI}}(t) = \mathbf{u}_{n,t}^T \mathbf{w}_{n,t}^{\text{FDI}}$. Therefore, it is found that the relation of $d_n^m(t)$ and $\mathbf{w}_{n,t}^m$ will be

$$d_n^m(t) - \mathbf{u}_{n,t}^T \mathbf{w}_{n,t}^{\text{FDI}} = d_n(t) - \mathbf{u}_{n,t}^T \mathbf{w}_{n,t}. \quad (38)$$

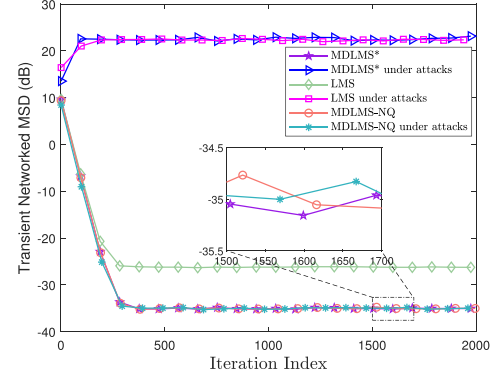


Fig. 2. Networked MSD of MDLMS*, LMS, and MDLMS-NQ under attacks-free and attacks.

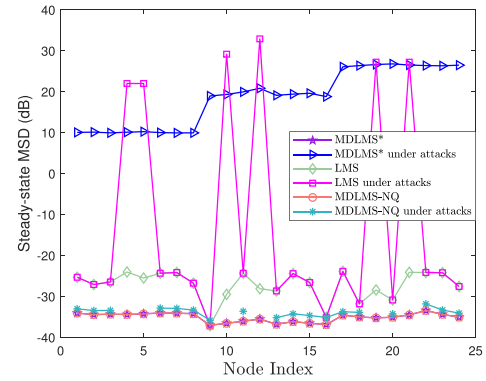


Fig. 3. Node MSD of MDLMS*, LMS, and MDLMS-NQ under attacks-free and attacks.

Given that these attacks can damage different types of data, they are considered in all experiments to verify the security and estimation performance of the proposed algorithm.

B. Experiments

To ensure uniformity in all experiments, the following parameters are set unless otherwise specified. The network topology is assumed to be as shown in Fig. 1 and has three clusters. For node n , the following parameters would be set uniformly as $\{\mu_n, M, \alpha', z, \gamma, \nu\} = \{0.02, 2, 0.3, 0.2, 0.5, 0.5\}$. The variances $\sigma_{v,n}^2$ of noise $v_n(t)$ are randomly obtained within (0,0.2), and the variances of signal $\mathbf{u}_{n,t}$ are uniformly set from 0.95 to 1.05. Besides, known cluster information is assumed specially to be known to a multi-task DLMS algorithm, denoted as MDLMS*, to provide a reference. In all experiments, the following nodes are selected to be attacked: nodes 10 and 12 invaded by Byzantine attack, nodes 19 and 21 invaded by the compromised communication attack, and nodes 4 and 5 invaded by FDI attack. Furthermore, all results are averaged over 300 trials. Note that the above cluster and attack information is unknown to all nodes and algorithms.

Figs. 2 and 3 examine the estimation performance when the network is invaded by all attacks. From these two figures, it is obvious that although the MDLMS* algorithm has known the cluster information, it cannot obtain the accurate estimation

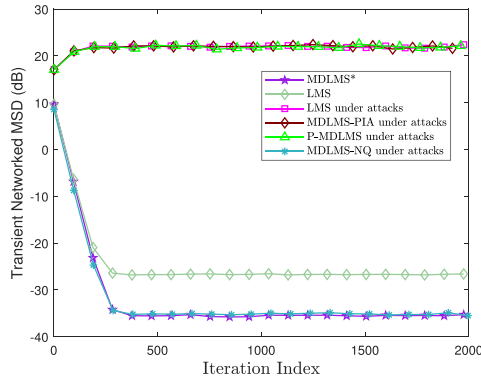


Fig. 4. Networked MSD of MDLMS-NQ and related multi-task algorithms with cluster learning under attacks.

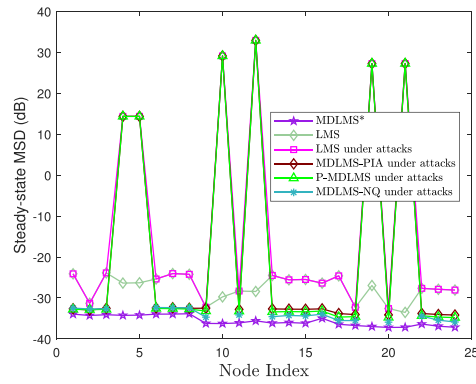


Fig. 5. Node MSD of MDLMS-NQ and related multi-task algorithms with cluster learning under attacks.

because of the lack of a secure mechanism. Observing LMS algorithm under attacks, it would not diffuse the impact of attacks due to the non-cooperation strategy, but its estimation performance also decreases significantly. In contrast, the proposed MDLMS-NQ algorithm achieves better estimation performance and still remains stable estimation behavior under attacks, which means that the designed strategy can learn cluster information and isolate the malicious nodes. Therefore, the proposed algorithm can work effectively without prior cluster information, even in the presence of attacks.

Fig. 4 shows the performance of various multi-task estimation algorithms with cluster learning strategy for the networked MSD under attacks, which include MDLMS*, LMS, LMS under attacks, MDLMS-PIA under attacks, P-MDLMS under attacks, MDLMS-NQ under attacks, and Fig. 5 depicts node MSD of them. This experiment verifies the cluster learning performance under attacks. From Fig. 4, even if MDLMS-PIA and P-MDLMS algorithms own the cluster learning strategy, their estimation performances degrade apparently under three attacks. Observing Fig. 5, it is clear that MDLMS-PIA and P-MDLMS algorithms can identify all differences among nodes, but there is no secure mechanism to obtain the resilience. From two figures, it is found that the proposed MDLMS-NQ algorithm still remains the same estimation performance.

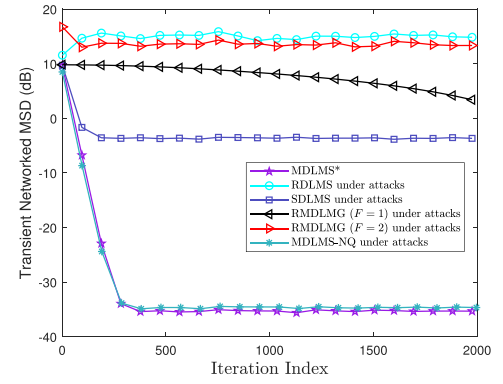


Fig. 6. Networked MSD of MDLMS-NQ and related secure estimation algorithms under attacks.

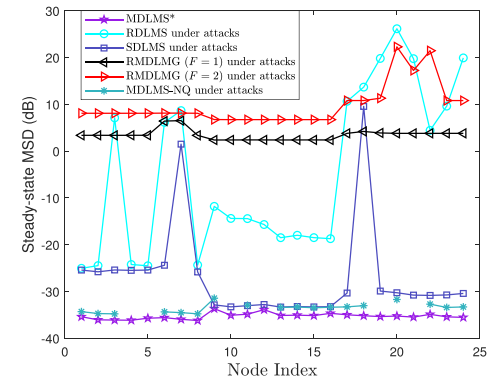


Fig. 7. Node MSD of MDLMS-NQ and related secure estimation algorithms under attacks.

Fig. 6 evaluates the performance of various secure algorithms for the networked MSD under different attacks, including MDLMS*, RDLMS under attacks, SDLMS under attacks, RMDLMG ($F = 1$) under attacks, RMDLMG ($F = 2$) under attacks, MDLMS-NQ under attacks. Fig. 7 shows node MSD of these algorithms. For RDLMS and SDLMS algorithms, they can distinguish the malicious nodes in single-task network, but they fail in multi-task network under attacks. In multi-task network, these two algorithms not only fail to estimate the unknown parameters, but all normal nodes have also experienced performance degradation. For the RMDLMG algorithm, it is necessary to set the number of outliers, namely, F , for all nodes. From Fig. 1, it is found that some normal nodes have one attacked neighbor, and some have two attacked neighbors. Therefore, it is impractical to predefine the parameter F for accurate estimation of w_n^o , making RMDLMG algorithm non-adaptive. If there is prior information about attacks, RMDLMG algorithm may achieve better estimation performance. But this prior information cannot be obtained in advance for either the network or the algorithm. Across all experiments, the proposed MDLMS-NQ algorithm achieves the best estimation performance, approaching that of the MDLMS* algorithm. Noting Fig. 1, it is observed that when some nodes are attacked, many normal border nodes are not f -local node while still satisfying f -model. Therefore, some secure algorithms fail to estimate the correct parameters. But based on the designed method and mechanism, as long as

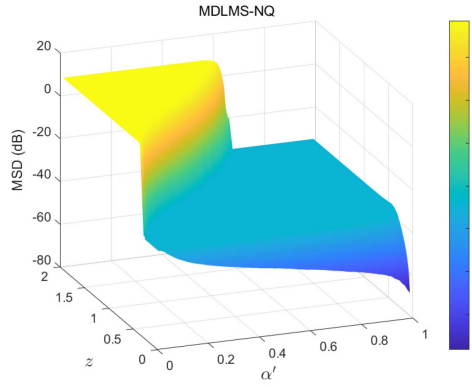


Fig. 8. MSD of MDLMS-NQ with different thresholds α' and different adjusting factors z .

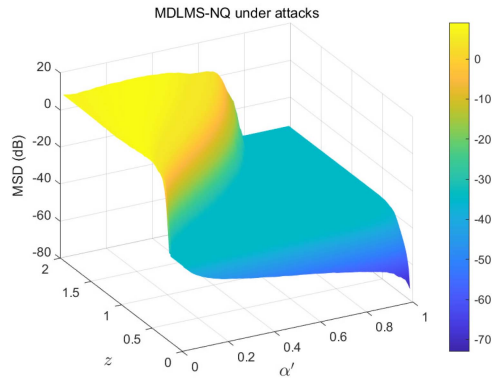


Fig. 9. MSD of MDLMS-NQ under attacks with different thresholds α' and different adjusting factors z .

all normal nodes in the multi-target estimation network are the reachable node, the resilience and learning capability of these nodes can be guaranteed. Consequently, the proposed algorithm is better suited to practical applications.

Figs. 8 and 9 investigate the estimation performance of MDLMS-NQ algorithm under different thresholds α' and different adjusting factors z . Through the above statements, z can adjust the degree of similarity between nodes n and j by formula (8). From Figs. 8 and 9, it is clear that to obtain the better detection, z should be (0,1), and the smaller the better. When z is closer to 0, the $g_{n,j}(t)$ of (8) will be closer to 1. Two nodes in the same cluster are similar to each other, so the threshold α' will approximate 1. From two figures, regardless of whether the proposed MDLMS-NQ algorithm is under attacks, to achieve the best estimation performance, z should be designed smaller and α' should be set larger within their certain suitable ranges.

Figs. 10 and 11 illustrate the impacts of various test thresholds γ and combination coefficients ν . From Fig. 10, the distributed network is secure, and all nodes are normal. Therefore, there are only differences among different clusters. In this situation, different clusters estimate different parameters, which means that the differences would be clear. Hence, the MDLMS-NQ algorithm maintains the same estimation performance under different γ and ν . From Fig. 11, when the distributed network is under three attacks, there are two differences due to the different

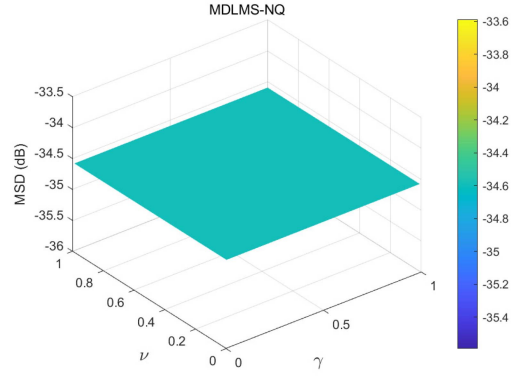


Fig. 10. MSD of MDLMS-NQ with different test thresholds γ and different combination coefficients ν .

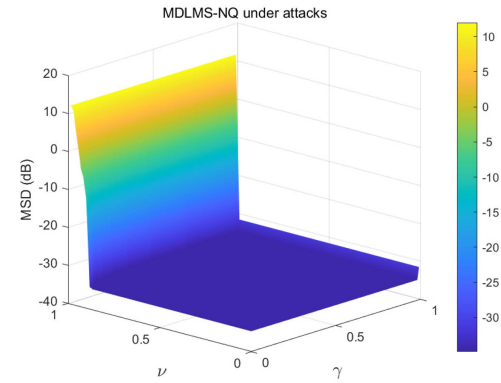


Fig. 11. MSD of MDLMS-NQ under attacks with different test thresholds γ and different combination coefficients ν .

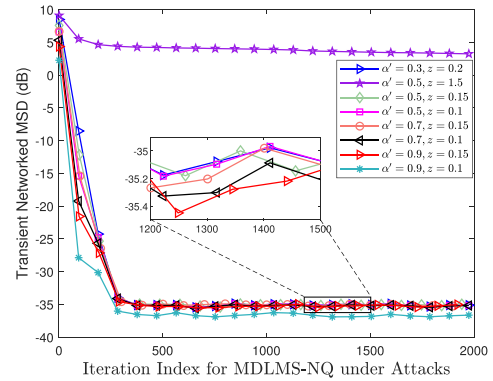


Fig. 12. Networked MSD under different α' and z .

clusters, and the normal and malicious nodes. Therefore, the parameters γ and ν will affect the estimation performance. Once γ and ν are set appropriately under attacks, the estimation performance of MDLMS-NQ algorithm will remain optimal. This is because, from (12) and (13), when $Q_n(t)$ can remember the behavior of the node, the designed node queue strategy will enable all normal nodes to achieve resilience and cluster learning.

From Figs. 8 and 9, it is found that the proposed MDLMS-NQ algorithm can obtain the better estimation performance under suitable parameters α' and z , so Fig. 12 further investigates the

situation. From Fig. 12, the proposed algorithm can achieve the best estimation performance under $\alpha' = 0.9$ and $z = 0.1$. But when α' and z are extremely close to 1 and 0, respectively, all nodes may isolate each other, causing the proposed algorithm to malfunction. Therefore, they are set in their certain suitable ranges. Besides, for $\alpha' = 0.5$ with different z values, it is clear that the coefficient z can be easily set. From all experiments, it can be concluded that the designed node queue strategy can make all reachable nodes achieve resilience, and the designed method can learn the cluster information under attacks. Moreover, compared with all related algorithms, the proposed MDLMS-NQ algorithm can obtain the best secure estimation performance without any prior information, which means that the proposed algorithm is more effective and efficient.

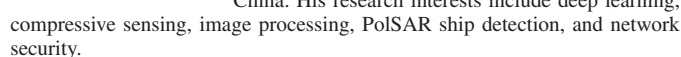
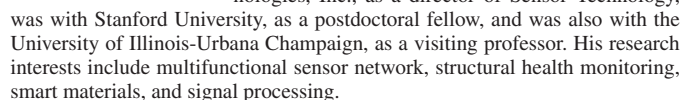
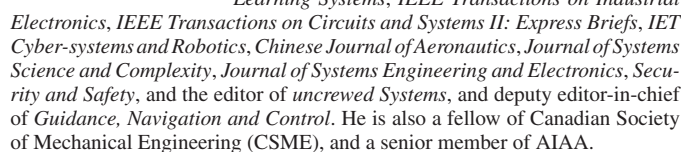
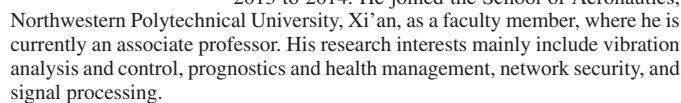
VI. CONCLUSION

In this paper, a resilient distributed MDLMS-NQ algorithm with node queue strategy is proposed to learn cluster information and obtain secure distributed estimation. In the proposed algorithm, an adaptive detection mechanism, which requires no prior network information, and a node queue strategy are designed to enable all normal nodes to obtain resilience and cluster learning. It is found that although f -local node can ensure the resilience of a node, the condition that all networked nodes are f -local node is not suitable for a multi-task network. Within the designed node queue strategy, a more compatible concept ‘*reachable node*’ is introduced for multi-task estimation network. The proposed reachable node and secure strategy better align with real-world scenarios, proving highly effective without the need for any prior network information. Performance analyses indicate that the proposed algorithm can distinguish all differences and obtain convergence. Simulation results demonstrate the stability, effectiveness and parameter simplicity of the proposed MDLMS-NQ algorithm, confirming its superior adaptability.

REFERENCES

- [1] U. Demirhan and A. Alkhateeb, “Integrated sensing and communication for 6G: Ten key machine learning roles,” *IEEE Commun. Mag.*, vol. 61, no. 5, pp. 113–119, May 2023.
- [2] Z. Wu, H. Li, Y. Qian, Y. Hua, and H. Gan, “Poison-resilient anomaly detection: Mitigating poisoning attacks in semi-supervised encrypted traffic anomaly detection,” *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 5, pp. 4744–4757, Sep.–Oct. 2024.
- [3] X. Zhou et al., “Decentralized P2P federated learning for privacy-preserving and resilient mobile robotic systems,” *IEEE Wireless Commun.*, vol. 30, no. 2, pp. 82–89, Apr. 2023.
- [4] I. Behnke and H. Austad, “Real-time performance of industrial IoT communication technologies: A review,” *IEEE Internet Things J.*, vol. 11, no. 5, pp. 7399–7410, Mar. 2024.
- [5] M. Yang, I. Tjuawinata, K.-Y. Lam, T. Zhu, and J. Zhao, “Differentially private distributed frequency estimation,” *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 5, pp. 3910–3926, Oct. 2023.
- [6] F.-Y. Wang et al., “Transportation 5.0: The DAO to safe, secure, and sustainable intelligent transportation systems,” *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 10262–10278, Oct. 2023.
- [7] F. Wan, T. Ma, Y. Hua, B. Liao, and X. Qing, “Secure distributed estimation under Byzantine attack and manipulation attack,” *Eng. Appl. Artif. Intell.*, vol. 116, Nov. 2022, Art. no. 105384.
- [8] Y. Hua, F. Wan, H. Gan, Y. Zhang, and X. Qing, “Distributed estimation with cross-verification under false data-injection attacks,” *IEEE Trans. Cybern.*, vol. 53, no. 9, pp. 5840–5853, Sep. 2023.
- [9] C. Seguin, O. Sporns, and A. Zalesky, “Brain network communication: Concepts, models and applications,” *Nature Rev. Neurosci.*, vol. 24, no. 9, pp. 557–574, Jul. 2023.
- [10] H. Shen, Y. Wang, J. Wu, J. H. Park, and J. Wang, “Secure control for Markov jump cyber-physical systems subject to malicious attacks: A resilient hybrid learning scheme,” *IEEE Trans. Cybern.*, vol. 54, no. 11, pp. 7068–7079, Nov. 2024.
- [11] S. M. R. H. Shawon, X. Liang, and M. Janbakhsh, “Optimal placement of distributed generation units for microgrid planning in distribution networks,” *IEEE Trans. Ind. Appl.*, vol. 59, no. 3, pp. 2785–2795, Jun. 2023.
- [12] F. S. Cattivelli and A. H. Sayed, “Diffusion LMS strategies for distributed estimation,” *IEEE Trans. Signal Process.*, vol. 58, no. 3, pp. 1035–1048, Mar. 2009.
- [13] Y. Hua, F. Wan, B. Liao, Y. Zong, S. Zhu, and X. Qing, “Adaptive multi-task clustering algorithm based on distributed diffusion least-mean-square estimation,” *Inf. Sci.*, vol. 606, pp. 628–648, Aug. 2022.
- [14] A. Almohammed, A. Zerguine, and M. Deriche, “A multitask incremental least mean square algorithm using orthonormal codes,” *Signal Process.*, vol. 222, Sep. 2024, Art. no. 109540.
- [15] Y. Yu, H. He, T. Yang, X. Wang, and R. C. de Lamare, “Diffusion normalized least mean M-estimate algorithms: Design and performance analysis,” *IEEE Trans. Signal Process.*, vol. 68, pp. 2199–2214, Mar. 2020.
- [16] F. Huang, S. Zhang, and W. X. Zheng, “Bayesian-learning-based diffusion least mean square algorithms over networks,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 13217–13231, Oct. 2024.
- [17] Y. Li et al., “Distributed hybrid-triggering-based secure dispatch approach for smart grid against DoS attacks,” *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 53, no. 6, pp. 3574–3587, Jun. 2023.
- [18] J. Li, W. Abbas, and X. Koutsoukos, “Resilient distributed diffusion in networks with adversaries,” *IEEE Trans. Signal Inf. Process. Netw.*, vol. 6, pp. 1–17, 2019.
- [19] T. Yu, R. C. de Lamare, and Y. Yu, “Robust resilient diffusion over multi-task networks against Byzantine attacks: Design, analysis and applications,” *IEEE Trans. Signal Process.*, vol. 70, pp. 2826–2841, 2022.
- [20] Y. Hua, F. Wan, B. Liao, S. Zhu, and X. Qing, “Resilient distributed information fusion under multiple malicious attacks,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 5, pp. 7367–7379, Oct. 2024.
- [21] Y. Liu and C. Li, “Secure distributed estimation over wireless sensor networks under attacks,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 54, no. 4, pp. 1815–1831, Aug. 2018.
- [22] H. Ma, Y. Xie, S. Tang, J. Hu, and X. Liu, “Threat-event detection for distributed networks based on spatiotemporal Markov random field,” *IEEE Trans. Dependable Secure Comput.*, vol. 19, no. 3, pp. 1735–1752, May/Jun. 2022.
- [23] Z. Khoshkalam, H. Zayyani, and M. Korki, “Secure multiple adaptive kernel diffusion LMS algorithm for distributed estimation over sensor networks,” *IET Wireless Sensor Syst.*, vol. 14, no. 6, pp. 477–483, Sep. 2024.
- [24] H. Zayyani and M. Korki, “Adaptive-width generalized correntropy diffusion algorithm for secure distributed estimation,” *IEEE Trans. Circuits Syst. II: Exp. Briefs*, vol. 70, no. 7, pp. 2700–2704, Jul. 2023.
- [25] J. Wang, D. Wang, H. Yan, and H. Shen, “Composite antidisturbance control for hidden Markov jump systems with multi-sensor against replay attacks,” *IEEE Trans. Autom. Control*, vol. 69, no. 3, pp. 1760–1766, Mar. 2023.
- [26] H. Zayyani, F. Oruji, and I. Fijalkow, “An adversary-resilient doubly compressed diffusion LMS algorithm for distributed estimation,” *Circuits, Syst., Signal Process.*, vol. 41, no. 11, pp. 6182–6205, Jul. 2022.
- [27] D. Yin, Y. Chen, R. Kannan, and P. Bartlett, “Byzantine-robust distributed learning: Towards optimal statistical rates,” in *Proc. Int. Conf. Mach. Learn.*, Jul. 2018, pp. 5650–5659.
- [28] H. Zayyani and M. Korki, “Secure distributed estimation via an average diffusion LMS and average likelihood ratio test,” *Digit. Signal Process.*, vol. 156, Jan. 2025, Art. no. 104782.
- [29] H. Zayyani, M. Salman, M. Rashdan, and H. Abuhilal, “Information-theoretic disturbance minimization for unified secure or robust distributed estimation in presence of impulsive FDI attack or impulsive noise,” in *IEEE Access*, vol. 13, pp. 163342–163353, 2025.
- [30] S. Marano and A. H. Sayed, “Decision learning and adaptation over multi-task networks,” *IEEE Trans. Signal Process.*, vol. 69, pp. 2873–2887, May 2021.
- [31] F. Hua, R. Nassif, C. Richard, H. Wang, and A. H. Sayed, “Online distributed learning over graphs with multitask graph-filter models,” *IEEE Trans. Signal Inf. Process. Netw.*, vol. 6, pp. 63–77, Jan. 2020.

- [32] T. Wang, Y. Li, F. Chen, and S. Duan, "Bayes-based distributed estimation in adversarial multitask networks," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 5, pp. 4004–4019, Oct. 2022.
- [33] R. Biswas and J. Wu, "Optimal filter assignment policy against distributed denial-of-service attack," *IEEE Trans. Dependable Secure Comput.*, vol. 19, no. 1, pp. 339–352, Feb. 2022.
- [34] F. Wan, Y. Hua, B. Liao, T. Ma, and X. Qing, "Distributed estimation with novel adaptive data selection based on a cross-matching mechanism," *Circuits, Syst., Signal Process.*, vol. 42, no. 10, pp. 6324–6346, Jun. 2023.
- [35] Y. Hua, F. Wan, H. Gan, and B. Liao, "One-step asynchronous data fusion DLMS algorithm," *IEEE Commun. Lett.*, vol. 25, no. 5, pp. 1660–1664, May 2021.
- [36] J. Chen, C. Richard, and A. H. Sayed, "Multitask diffusion adaptation over networks," *IEEE Trans. Signal Process.*, vol. 62, no. 16, pp. 4129–4144, Aug. 2014.
- [37] H. Yuan, X. Zhao, and L. Yu, "A distributed Bayesian algorithm for data fault detection in wireless sensor networks," in *Proc. Int. Conf. Inf. Netw.*, Jan. 2015, pp. 63–68.
- [38] A. Mitra and S. Sundaram, "Secure distributed observers for a class of linear time invariant systems in the presence of Byzantine adversaries," in *Proc. IEEE 55th Conf. Decis. Control*, Dec. 2016, pp. 2709–2714.
- [39] M. Doostmohammadian and M. Pirani, "On the design of resilient distributed single time-scale estimators: A graph-theoretic approach," *IEEE Trans. Netw. Sci. Eng.*, vol. 12, no. 3, pp. 3975–3984, Oct. 2025.
- [40] M. Doostmohammadian, H. Zarrabi, H. R. Rabiee, U. A. Khan, and T. Charalambous, "Distributed detection and mitigation of biasing attacks over multi-agent networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 4, pp. 3465–3477, Oct./Dec. 2021.
- [41] F. D. Mandanas and C. L. Kotropoulos, "Robust multidimensional scaling using a maximum correntropy criterion," *IEEE Trans. Signal Process.*, vol. 65, no. 4, pp. 919–932, Feb. 2017.
- [42] Y. Hua, F. Chen, S. Deng, S. Duan, and L. Wang, "Secure distributed estimation against false data injection attack," *Inf. Sci.*, vol. 515, pp. 248–262, Apr. 2020.
- [43] C. Liu, X. Hu, X. Chen, J. Wei, and W. Liu, "SDIM: A subtly designed invertible matrix for enhanced privacy-preserving outsourcing matrix multiplication and related tasks," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 4, pp. 3469–3486, Jul./Aug. 2023.
- [44] Y. Hua, H. Gan, F. Wan, X. Qing, and F. Liu, "Distributed estimation with adaptive cluster learning over asynchronous data fusion," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 5, pp. 5262–5274, Oct. 2023.
- [45] G. Lu, W. Chen, and Q. Huang, "Distributed diffusion least mean square algorithm based on the reputation mechanism," *J. Electron. Inf. Technol.*, vol. 37, no. 5, pp. 1234–1240, May 2015.
- [46] C. Quan, S. Bulusu, B. Geng, Y. S. Han, N. Sriranga, and P. K. Varshney, "On ordered transmission based distributed Gaussian shift-in-mean detection under Byzantine attacks," *IEEE Trans. Signal Process.*, vol. 71, pp. 3343–3356, 2023.
- [47] L. Zhang, A. A. Ding, Y. Fei, and P. Luo, "Efficient nonprofiling 2nd-order power analysis on masked devices utilizing multiple leakage points," *IEEE Trans. Dependable Secure Comput.*, vol. 16, no. 5, pp. 843–855, Sep./Oct. 2019.
- [48] H. Wu, B. Zhou, and C. Zhang, "Secure distributed estimation against data integrity attacks in internet-of-things systems," *IEEE Trans. Automat. Sci. Eng.*, vol. 19, no. 3, pp. 2552–2565, Jul. 2022.



Youmin Zhang (Fellow, IEEE) is currently a professor with the Department of Mechanical, Industrial and Aerospace Engineering and also with the Concordia Institute of Aerospace Design and Innovation (CIADI), Concordia University, Canada. His research interests mainly include the areas of condition monitoring, fault diagnosis and fault-tolerant control systems; detection, and protection by combining with remote sensing techniques. He is also an associate editor for *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Industrial Electronics*, *IEEE Transactions on Circuits and Systems II: Express Briefs*, *IET Electronics*, *Chinese Journal of Aeronautics*, *Journal of Systems Engineering*, *Journal of Systems Engineering and Electronics*, *Security and Control*. He is also a fellow of Canadian Society of Mechanical Engineers (CSME), and a senior member of AIAA.

Xinlin Qing (Member, IEEE) received the MSc degrees in experimental mechanics from Tianjin University, Tianjin, China, in 1991, and the PhD degree in solid mechanics from Tsinghua University, Beijing, China, in 1994. He is currently a distinguished professor with the College of Aerospace Engineering, Xiamen University. Before he joined Xiamen University, he was with the Beijing Aeronautical Science and Technology Research Institute of COMAC, as a senior technical fellow, was with Accell Technologies, Inc., as a director of Sensor Technology, as a postdoctoral fellow, and was also with the Bana Champaign, as a visiting professor. His research interests include sensor network, structural health monitoring, and signal processing.

Hongping Gan (Member, IEEE) received the PhD degree in communication and information engineering from the State Key Laboratory of ISN, Xidian University, Xi'an, China, in 2020. From 2018 to 2019, he was with the School of Information Technology and Electrical Engineering, The University of Queensland, Brisbane, Australia, as a visiting scholar sponsored by the China Scholarship Council. He is currently an associate professor with the School of Software, Northwestern Polytechnical University, China. His research interests include deep learning.



Yi Hua received the PhD degree in aerospace science and technology with the School of Aeronautics, Northwestern Polytechnical University, Xi'an, China, in 2025. From 2023 to 2025, he was with the School of Electrical Engineering and Computer Science, The University of Queensland, Brisbane, QLD, Australia, as a visiting scholar sponsored by the China Scholarship Council. He is currently with the College Information and Artificial Intelligence, Yangzhou University, Yangzhou, China. His research interests include distributed signal processing, net-

network security, image processing, graph signal processing, and data fusion of multiple aircraft.